

Problem 2: Bloch sphere

1- Given

$$A = \frac{1}{2} (r_0 I + \vec{r} \cdot \vec{\sigma})$$

Required

Show that $A^\dagger = A$

Solution

$$A = \frac{1}{2} (r_0 I + \vec{r} \cdot \vec{\sigma})$$

$$= \frac{1}{2} (r_0 I + r_1 \sigma_1 + r_2 \sigma_2 + r_3 \sigma_3)$$

$$= \frac{1}{2} \left[r_0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + r_1 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + r_2 \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} + r_3 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right]$$

$$= \frac{1}{2} \left(\begin{pmatrix} r_0 & 0 \\ 0 & r_0 \end{pmatrix} + \begin{pmatrix} 0 & r_1 \\ r_1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -ir_2 \\ ir_2 & 0 \end{pmatrix} + \begin{pmatrix} r_3 & 0 \\ 0 & -r_3 \end{pmatrix} \right)$$

$$= \frac{1}{2} \begin{pmatrix} r_0 + r_3 & r_1 - ir_2 \\ r_1 + ir_2 & r_0 - r_3 \end{pmatrix}$$

Then calculate $A^\dagger$

$$A^\dagger = (A^x)^T$$

$$= \left(\frac{1}{2} \begin{pmatrix} r_0 + r_3 & r_1 - ir_2 \\ r_1 + ir_2 & r_0 - r_3 \end{pmatrix} \right)^T$$

$$= \left(\frac{1}{2} \begin{pmatrix} r_0 + r_3 & r_1 + ir_2 \\ r_1 - ir_2 & r_0 - r_3 \end{pmatrix} \right)^T$$

$$= \frac{1}{2} \begin{pmatrix} r_0 + r_3 & r_1 - ir_2 \\ r_1 + ir_2 & r_0 - r_3 \end{pmatrix}$$

$\underbrace{\hspace{10em}}_A$

$A^\dagger = A$, Therefore any operators written 2)

In the form of

$A = \frac{1}{2}(\gamma_0 I + \vec{\gamma} \cdot \vec{\sigma})$ is Hermitian operator.

② Show that the operator β is given by

$$\beta = \frac{1}{2}(I + \vec{\gamma} \cdot \vec{\sigma})$$

Given

$$\beta = \frac{1}{2}(I + \vec{\gamma} \cdot \vec{\sigma})$$

Required

Hermitian
Unit trace
Positivity

Show all this for the
validity of density matrix

Solution

1. Hermiticity, $\beta^\dagger = \beta$ - Show this?

$$\beta^\dagger = (\beta)^\dagger$$

$$= \left[\frac{1}{2}(I + \vec{\gamma} \cdot \vec{\sigma}) \right]^\dagger$$

$$= \frac{1}{2} \left(I^\dagger + (\vec{\gamma} \cdot \vec{\sigma})^\dagger \right)$$

$$= \frac{1}{2} \left(I + \left(\sum_i \gamma_i \sigma_i \right)^\dagger \right)$$

$$= \frac{1}{2} \left(I + \sum_i \gamma_i^* \sigma_i^\dagger \right)$$

$$= \frac{1}{2} (I + \vec{\gamma} \cdot \vec{\sigma})$$

$\underbrace{\hspace{10em}}_{= \beta}$

$$\underline{\underline{\beta^\dagger = \beta}}$$

Where

$$I^\dagger = I$$

$$\begin{aligned} (\vec{\gamma} \cdot \vec{\sigma})^\dagger &= \left(\sum_i \gamma_i \sigma_i \right)^\dagger \\ &= \sum_i \gamma_i^* \sigma_i^\dagger \end{aligned}$$

$$\left\{ \begin{aligned} \sigma_i^\dagger &= \sigma_i \\ \gamma_i^* &= \gamma_i \end{aligned} \right. \quad \begin{aligned} (\vec{\gamma} \cdot \vec{\sigma})^\dagger &= \sum_i \gamma_i \sigma_i \\ &= \vec{\gamma} \cdot \vec{\sigma} \\ &\underline{\underline{=}} \end{aligned}$$

c. Unit trace

$$\begin{aligned}
 \text{tr}(S) &= \text{tr}\left(\frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})\right) \\
 &= \frac{1}{2}(\text{tr}(I) + \vec{r} \cdot \text{tr}(\vec{\sigma})) \\
 &= \frac{1}{2}(2 + \vec{r} \cdot 0) \\
 &= \frac{1}{2}(2 + 0) \\
 &= \frac{2}{2} \\
 &= \underline{\underline{1}}
 \end{aligned}$$

Where

$$\begin{aligned}
 \text{tr}(I) &= 2 \\
 \text{tr}(I_{2 \times 2}) &= 2 \\
 \text{tr}(\vec{\sigma}) &= 0
 \end{aligned}$$

Positivity

The eigenvalues of a 2×2 matrix

$$A = aI + b, \text{ where } a \text{ and } b \text{ are real, } a \pm \|b\|$$

so

$$\begin{aligned}
 S &= \frac{1}{2}I + \frac{1}{2}(\vec{r} \cdot \vec{\sigma}), \quad a = \frac{1}{2} \\
 b &= \frac{1}{2}\vec{r}
 \end{aligned}$$

$$\lambda_{\pm} = \frac{1}{2} \pm \|\frac{1}{2}\vec{r}\| = \frac{1}{2} \pm \frac{1}{2}\|\vec{r}\|$$

$$\lambda_{\pm} = \frac{1}{2} \pm \frac{1}{2}\|\vec{r}\|$$

for, λ_{+}

$$\lambda_{+} = \frac{1}{2} + \frac{1}{2}\|\vec{r}\|$$

for λ_{-}

$$\lambda_{-} = \frac{1}{2} - \frac{1}{2}\|\vec{r}\|$$

For $\lambda_{-} \geq 0$, we require

$$\frac{1}{2} - \frac{1}{2}\|\vec{r}\| \geq 0$$

$$\frac{1}{2} - \frac{1}{2}\|\vec{r}\| \geq 0$$

$$-\frac{1}{2}\|\vec{r}\| \geq 0 - \frac{1}{2}$$

$$-\frac{1}{2}\|\vec{r}\| \geq -\frac{1}{2}$$

$$\|\vec{r}\| \leq 1$$

3 Required

Show that:

$$\text{PurValue}(S) = \text{Tr}(S^2) = \frac{1}{2}(2 + |\vec{r}|^2)$$

Solution

$$S = \frac{1}{2}(\mathbf{I} + \vec{r} \cdot \vec{\sigma})$$

Compute

$$S^2 = \left(\frac{1}{2}(\mathbf{I} + \vec{r} \cdot \vec{\sigma}) \right)^2$$

$$= \frac{1}{4}(\mathbf{I} + \vec{r} \cdot \vec{\sigma})(\mathbf{I} + \vec{r} \cdot \vec{\sigma})$$

$$= \frac{1}{4}(\mathbf{I}^2 + \vec{r} \cdot \vec{\sigma} \mathbf{I} + \vec{r} \cdot \vec{\sigma} \mathbf{I} + (\vec{r} \cdot \vec{\sigma})(\vec{r} \cdot \vec{\sigma}))$$

$$= \frac{1}{4}(\mathbf{I}^2 + 2\vec{r} \cdot \vec{\sigma} \mathbf{I} + |\vec{r}|^2 \mathbf{I}^2)$$

$$= \frac{1}{4}(\mathbf{I}^2 + 2\vec{r} \cdot \vec{\sigma} \mathbf{I} + |\vec{r}|^2 \mathbf{I})$$

$$= \frac{1}{4}(\mathbf{I} + 2\vec{r} \cdot \vec{\sigma} + |\vec{r}|^2 \mathbf{I})$$

Take the trace of S^2

$$\text{PurValue}(S) = \text{Tr}(S^2)$$

$$= \text{Tr}\left(\frac{1}{4}(\mathbf{I} + 2\vec{r} \cdot \vec{\sigma} + |\vec{r}|^2 \mathbf{I})\right)$$

$$= \frac{1}{4}(\text{Tr}(\mathbf{I}) + 2\vec{r} \cdot \text{Tr}(\vec{\sigma}) + |\vec{r}|^2 \text{Tr}(\mathbf{I}))$$

$$= \frac{1}{4}(2 + 2\vec{r} \cdot 0 + |\vec{r}|^2 \cdot 2)$$

$$= \frac{1}{4}(2 + 2|\vec{r}|^2)$$

$$\begin{aligned} \text{Tr}(\rho^2) &= \frac{2}{4} (2 + |\psi|^2) \\ &= \frac{1}{2} (2 + |\psi|^2) \end{aligned}$$

5)

The Bloch vector must be a $|\psi| = 2$

$$\begin{aligned} \text{Tr}(\rho^2) &= \frac{1}{2} (2 + |\psi|^2) \\ &= \frac{1}{2} (2 + 2) \\ &= \frac{2}{2} \\ &= \underline{\underline{1}} \end{aligned}$$

Purity = 1 for pure state,
 $|\psi| = 2$

Problem 2: Operator-sum representation

2) Solution

Show that the action of the unitary on the system can be expressed as

$$U(\rho \otimes |0\rangle\langle 0|_{env})U^\dagger = \sum_{k,l} E_k \rho E_l^\dagger \otimes |k\rangle\langle l|_{env}$$

Therefore,

We define $E_k = (I \otimes \langle k|)(I \otimes |0\rangle)$ and check

$$\begin{aligned} \sum_{k,l} E_k \rho E_l^\dagger \otimes |k\rangle\langle l| &= \sum_{k,l} (I \otimes \langle k|) U (I \otimes |0\rangle) \rho (I \otimes \langle l|) U^\dagger (I \otimes |0\rangle) \otimes |k\rangle\langle l| \\ &= \sum_{k,l} (I \otimes \langle k|) U (I \otimes |0\rangle) \rho (I \otimes \langle l|) U^\dagger (I \otimes |0\rangle) \otimes |k\rangle\langle l| \end{aligned}$$

5)

$$= \sum_{k_1} (I \otimes L_{k_1}) U (S \otimes |0\rangle_{L0}) U^\dagger (I \otimes |k\rangle_{L1}) \otimes |k\rangle_{L1}$$

$$= U (S \otimes |0\rangle_{L0}) U^\dagger$$

where

$$\langle i, m | \left(\sum_{k_1} (I \otimes L_{k_1}) U (S \otimes |0\rangle_{L0}) U^\dagger (I \otimes |k\rangle_{L1}) \otimes |k\rangle_{L1} \right) | j, n \rangle$$

$$= (\langle i | \otimes I) \left(\sum_{k_1} (I \otimes L_{k_1}) U (S \otimes |0\rangle_{L0}) U^\dagger (I \otimes |k\rangle_{L1}) \delta_{k_1, m} \delta_{k_1, n} \right) (|j\rangle \otimes I)$$

$$= (\langle i | \otimes I) (I \otimes L_{m |}) U (S \otimes |0\rangle_{L0}) U^\dagger (I \otimes |n\rangle_{L1}) (|j\rangle \otimes I)$$

$$= \langle i, m | U (S \otimes |0\rangle_{L0}) U^\dagger | j, n \rangle$$

2) Given

$$E_k = \{ |k\rangle_{env} \} \cup |0\rangle_{env}$$

$$k = (0, h)$$

$$p_i = |i\rangle_{L1}, p_{00}, p = 0, h$$

Initial state: $|1\rangle_{L1} \otimes |0\rangle_{env} = |0\rangle_{env}$

Solution

The interaction is given by the unitary transformation

$$U = p_0 \otimes I_{env} + p_h \otimes X_{env}$$

Initial state $|0\rangle$

$$E_0 = |0\rangle \otimes |0\rangle$$

$$= |0\rangle (p_0 \otimes I_{env} + p_h \otimes X_{env}) |0\rangle$$

$$= |0\rangle (p_0 \otimes I_{env}) |0\rangle + |0\rangle (p_h \otimes X_{env}) |0\rangle$$

$$= p_0 \langle 0 | I_{env} | 0 \rangle + p_1 \langle 0 | X_{env} | 0 \rangle$$

$$= p_0 \underbrace{\langle 0 | 0 \rangle}_{=1} + p_1 \underbrace{\langle 0 | 1 \rangle}_{=0}$$

$$= p_0 \cdot 1 + p_1 \cdot 0$$

$$= \underline{\underline{p_0}}$$

$$\text{Given, } p_0 = |0\rangle\langle 0|$$

$$\underline{\underline{E_0 = p_0 = |0\rangle\langle 0|}}$$

$$E_1 = |1\rangle\langle 1|$$

$$= |1\rangle\langle 1| (p_0 \otimes I_{env} + p_1 \otimes X_{env}) |0\rangle$$

$$= |1\rangle\langle 1| p_0 I_{env} |0\rangle + |1\rangle\langle 1| p_1 X_{env} |0\rangle$$

$$= p_0 |1\rangle\langle 1| I_{env} |0\rangle + p_1 |1\rangle\langle 1| X_{env} |0\rangle$$

$$= p_0 \underbrace{\langle 1 | 0 \rangle}_{=0} + p_1 \underbrace{\langle 1 | 1 \rangle}_{=1}$$

$$= p_0 \cdot 0 + p_1 \cdot 1$$

$$= \underline{\underline{p_1}}$$

$$\underline{\underline{E_1 = p_1 = |1\rangle\langle 1|}}$$

Operator sum representation

Formula:

$$\hat{Z}(\rho) = \sum_k E_k \rho E_k^\dagger$$

$$\hat{Z}(\rho) = E_0 \rho E_0^\dagger + E_1 \rho E_1^\dagger$$

$$= P_0 \rho P_0^\dagger + P_1 \rho P_1^\dagger$$

$$= \underbrace{107401 \rho (107401)^\dagger}_{=107401} + \underbrace{117211 \rho (117211)^\dagger}_{=117211}$$

$$= 107401 \rho 107401 + 117211 \rho 117211$$

$$\underline{\underline{\quad \quad \quad // \quad \quad \quad}}$$

$$2) \text{ For } \underline{\underline{\quad}} = \frac{x}{\sqrt{2}} \otimes I_{\text{env}} + \frac{y}{2} \otimes X_{\text{env}}$$

Solution , Initial state $|0\rangle$

$$E_0 = 4010107$$

$$= 401 \left[\left(\frac{x}{\sqrt{2}} \otimes I_{\text{env}} \right) + \left(\frac{y}{2} \otimes X_{\text{env}} \right) \right] |0\rangle$$

$$= 401 \left(\left(\frac{x}{\sqrt{2}} \otimes I_{\text{env}} \right) |0\rangle + \left(\frac{y}{2} \otimes X_{\text{env}} \right) |0\rangle \right)$$

$$= \frac{x}{\sqrt{2}} 401 I_{\text{env}} |0\rangle_{\text{env}} + \frac{y}{2} 401 X_{\text{env}} |0\rangle_{\text{env}}$$

$$= \frac{x}{\sqrt{2}} 401 |0\rangle + \frac{y}{2} 401 |1\rangle$$

$$= \frac{x}{\sqrt{2}} //$$

$$E_0 = \frac{x}{\sqrt{2}}$$

$$E_1 = \frac{1}{\sqrt{2}}(x \otimes I_{env} + \frac{1}{\sqrt{2}}(x \otimes x_{env}))|0\rangle$$

$$= \frac{x}{\sqrt{2}} \frac{1}{\sqrt{2}}(I_{env}|0\rangle_{env} + \frac{1}{\sqrt{2}}(x_{env}|0\rangle_{env}))$$

$$= \frac{x}{\sqrt{2}} \frac{1}{\sqrt{2}}(I_{env}|0\rangle_{env} + \frac{1}{\sqrt{2}}(x_{env}|0\rangle_{env}))$$

$$= \frac{x}{\sqrt{2}} \frac{1}{\sqrt{2}}(I_{env}|0\rangle_{env} + \frac{1}{\sqrt{2}}(x_{env}|0\rangle_{env}))$$

$$= \frac{x}{\sqrt{2}} \cdot 0 + \frac{1}{\sqrt{2}} \cdot 1$$

$$= \frac{1}{\sqrt{2}}$$

Operator sum representation

$$\Sigma(S) = E_0 S E_0^\dagger + E_1 S E_1^\dagger$$

$$= \frac{x}{\sqrt{2}} S \left(\frac{x}{\sqrt{2}}\right)^\dagger + \frac{1}{\sqrt{2}} S \left(\frac{1}{\sqrt{2}}\right)^\dagger$$

$$= \frac{x}{\sqrt{2}} S \frac{x}{\sqrt{2}} + \frac{1}{\sqrt{2}} S \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2} x S x + \frac{1}{2} 1 S 1$$

$$= \frac{1}{2} (x S x + 1 S 1)$$

$$\underline{\underline{\quad \quad \quad}}$$

Where

$$(\sigma_i)^\dagger = \sigma_i \Rightarrow \begin{cases} (x)^\dagger = x \\ (-1)^\dagger = -1 \end{cases}$$

3 Given

$$S_{AB} = S'$$

$$S_{BC} = S' = \Sigma(S)$$

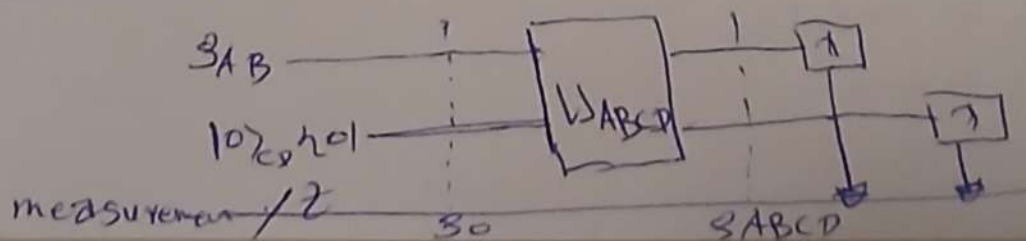
Required

$$\Sigma(S) = S'$$

$$\sum_k E_k^\dagger E_k = I$$

Show this

Solution - to solve the problem I used the quantum circuits



1. First calculate S_0

$$S_0 = S_{AB} \otimes |0\rangle_{CD} \langle 0|$$

2. Calculate S_{ABCD} by apply U_{ABCD} on S_0

$$\begin{aligned} S_{ABCD} &= U_{ABCD} (S_0) U_{ABCD}^\dagger \\ &= U_{ABCD} (S_{AB} \otimes |0\rangle_{CD} \langle 0|) U_{ABCD}^\dagger \end{aligned}$$

3. Calculate S_{BC} by tracing out (AD) from S_{ABCD}

$$\begin{aligned} S_{BC} &= \text{Tr}_{AD}(S_{ABCD}) \\ &= \text{Tr}_{AD}(U_{ABCD} (S_{AB} \otimes |0\rangle_{CD} \langle 0|) U_{ABCD}^\dagger) \end{aligned}$$

By using $\{|k\rangle_{AD}\}$ to perform the partial trace over AD

$$\begin{aligned} S_{BC} &= \sum_k \langle k|_{AD} (U_{ABCD} (S_{AB} \otimes |0\rangle_{CD} \langle 0|) U_{ABCD}^\dagger) |k\rangle_{AD} \\ &= \sum_k \left(\langle k|_{AD} U_{ABCD} (\cdot \otimes |0\rangle_{CD}) \right) S_{AB} \left(\left(\langle k|_{AD} \otimes \cdot \right) U_{ABCD}^\dagger \right) \\ &= \sum_k \underbrace{\left(\langle k|_{AD} U_{ABCD} (\cdot \otimes |0\rangle_{CD}) \right)}_{= E_k} S_{AB} \underbrace{\left(\left(\langle k|_{AD} \otimes \cdot \right) U_{ABCD}^\dagger \right)}_{=(E_k)^\dagger} \\ &= \sum_k E_k S_{AB} (E_k)^\dagger \\ &= \sum_k \underline{\underline{E_k S_{AB} E_k^\dagger}} \end{aligned}$$

Substitute all terms based on the problem

where

$$S_{AB} = S$$

$$S' = S_{BC} = S(S) \quad \left. \vphantom{S' = S_{BC} = S(S)} \right\} \text{given}$$

$$E(S) = \sum_k E_k S E_k^\dagger$$

=====

The solution satisfied the completeness $\sum_k E_k^\dagger E_k = I$

$$\sum_k E_k^\dagger E_k = \sum_k [(\langle 0 | \otimes \langle 0 |_{CD}) W^\dagger | k \rangle_{AD}] [\langle k |_{AD} W (\langle 0 | \otimes | 0 \rangle_{CD})]$$

$$= [(\langle 0 | \otimes \langle 0 |_{CD}) W^\dagger \underbrace{\left(\sum_k | k \rangle_{AD} \langle k | \right)}_{= I} W (\langle 0 | \otimes | 0 \rangle_{CD})]$$

$$= [(\langle 0 | \otimes \langle 0 |_{CD}) W^\dagger W \underbrace{(\langle 0 | \otimes | 0 \rangle_{CD})}_{= I}]$$

$$= [(\langle 0 | \otimes \langle 0 |_{CD}) (\langle 0 | \otimes | 0 \rangle_{CD})]$$

$$= \langle 0 | 0 \rangle_{CD}$$

$$= 1$$

=====

Problem 3: Quantum operation on qubits,

Given

$$S_1(S) = E X P X + (I - E) S$$

$$E_2(S) = \frac{E}{2} \text{Tr}[S] I + (I - E) S$$

$$E_2(S) = E \text{Tr}[P] \rho + (1-E)S$$

Required

- ✓ Trace-preserving map
 - ✓ completely positive
 - ✓ linear
- } Show this in an 8 noise channel

Solution

① Trace-preserving

$$\begin{aligned}
 \text{Tr}[E_1(S)] &= \text{Tr}[E X P X + (1-E)S] \\
 &= \text{Tr}[E X P X] + \text{Tr}[(1-E)S] \\
 &= E \text{Tr}(X P X) + (1-E) \text{Tr}(S) \\
 &= E \text{Tr}(X X P) + (1-E) \text{Tr}(S) \\
 &= E \text{Tr}(X^2 P) + (1-E) \text{Tr}(S) \\
 &= E \text{Tr}(I S) + (1-E) \text{Tr}(S) \\
 &= E \underbrace{\text{Tr}(S)}_{=1} + (1-E) \underbrace{\text{Tr}(S)}_{=1} \\
 &= E \cdot 1 + (1-E) \cdot 1 \\
 &= E + 1 - E \\
 &= 1 //
 \end{aligned}$$

$$\begin{aligned}
 \text{Tr}(E_2(S)) &= \text{Tr}\left(\frac{E}{2} \underbrace{\text{Tr}(S)}_{=1} I + (1-E)S\right) \\
 &= \frac{E}{2} \text{Tr}(I) + (1-E) \text{Tr}(S)
 \end{aligned}$$

$$= \frac{E}{2} \text{Tr}(\underline{I}) + (2-E) \text{Tr}(S)$$

$$= \frac{2 \cdot E}{2} + (2-E) \cdot 1$$

$$= \frac{2 \cdot E}{2} + 2 - E$$

$$= E + 2 - E$$

$$= 2 + E - E$$

$$= 2 + 0$$

$$= 2 //$$

$$\hat{\pi}(\Sigma_3(S)) = \text{Tr} \left[E \text{Tr}(\underline{S}) |0\rangle\langle 0| + (2-E)S \right]$$

$$= \text{Tr} [E \cdot 2 \cdot |0\rangle\langle 0| + (2-E)S]$$

$$= \text{Tr} [E(|0\rangle\langle 0|) + (2-E)S]$$

$$= E \text{Tr}(|0\rangle\langle 0|) + (2-E) \text{Tr}(S)$$

$$= E \underbrace{\text{Tr}(|0\rangle\langle 0|)}_{=2} + (2-E) \underbrace{\text{Tr}(S)}_{=2}$$

$$= E \cdot 2 + (2-E) \cdot 2$$

$$= E + 2 - E$$

$$= 2 + E - E$$

$$= 2 + 0$$

$$= 2 //$$

⊗ Complete positivity

$$\Sigma_A(S) = E X S X + (2-E)S$$

Choose Kraus operators

$$K_0 = \sqrt{\lambda - E} I$$

$$K_1 = \sqrt{E} X$$

Then,

$$\begin{aligned} \sum_{i=0}^d K_i^\dagger \rho K_i &= K_0^\dagger \rho K_0 + K_1^\dagger \rho K_1 \\ &= \sqrt{\lambda - E} \rho \sqrt{\lambda - E} + \sqrt{E} X \rho \sqrt{E} X^\dagger \\ &= (\sqrt{\lambda - E})^2 \rho + (\sqrt{E})^2 X \rho X^\dagger \\ &= (\lambda - E) \rho + E X \rho X^\dagger, \\ &= (\lambda - E) \rho + E X \rho X = \Sigma_\lambda(\rho) \end{aligned}$$

$$\sum_{i=0}^d K_i^\dagger \rho K_i = (\lambda - E) \rho + E X \rho X = \Sigma_\lambda(\rho)$$

$$\sum_{i=0}^d K_i^\dagger \rho K_i = \Sigma_\lambda(\rho)$$

Since Σ_λ is written as $\sum_i K_i^\dagger \rho K_i$ it is CP

For, $\Sigma_\lambda(\rho) = \frac{E}{2} \text{Tr}(\rho) I + (\lambda - E) \rho$

Choose Kraus operators

$$K_0 = \sqrt{\lambda - \frac{3E}{4}} I$$

$$K_2 = \sqrt{\frac{E}{4}} \sigma_i, \text{ where } (i = 1, 2, 3)$$

given

$$\begin{aligned}
 \sum_{i=0}^3 K_i^0 S K_i^{+T} &= \sqrt{1 - \frac{3E}{4}} S \sqrt{1 - \frac{3E}{4}}^T + \sqrt{\frac{E}{4}} \sigma_i^0 S \left(\sqrt{\frac{E}{4}} \sigma_i^0 \right)^T \\
 &= \left(\sqrt{1 - \frac{3E}{4}} \right)^2 S + \left(\sqrt{\frac{E}{4}} \right)^2 \sigma_i^0 S \underbrace{(\sigma_i^0)^T}_{= \sigma_i^0} \\
 &= \left(1 - \frac{3E}{4} \right) S + \frac{E}{4} \sigma_i^0 S \sigma_i^0 \\
 &= \left(1 - \frac{3E}{4} \right) S + \frac{E}{4} \left[\sigma_1^0 S \sigma_1^0 + \sigma_2^0 S \sigma_2^0 + \sigma_3^0 S \sigma_3^0 \right] \\
 &\quad \quad \quad = \sum_{i=1}^3 \sigma_i^0 S \sigma_i^0 \\
 &= \left(1 - \frac{3E}{4} \right) S + \frac{E}{4} \sum_{i=1}^3 \sigma_i^0 S \sigma_i^0
 \end{aligned}$$

Subst Identity

$$\boxed{\sum_{i=1}^3 \sigma_i^0 S \sigma_i^0 = 2I - S}$$

$$\begin{aligned}
 \sum_{i=0}^3 K_i^0 S K_i^{+T} &= \left(1 - \frac{3E}{4} \right) S + \frac{E}{4} \underbrace{\sum_{i=1}^3 \sigma_i^0 S \sigma_i^0}_{= 2I - S} \\
 &= 2I - S
 \end{aligned}$$

$$\begin{aligned}
 \sum_{i=0}^3 K_i^0 S K_i^{+T} &= \left(1 - \frac{3E}{4} \right) S + \frac{E}{4} (2I - S) \\
 &= S - \frac{3SE}{4} + \frac{2EI}{4} - \frac{SE}{4}
 \end{aligned}$$

$$\begin{aligned}
\sum_{i=0}^3 K_i^\dagger \rho K_i &= \rho - \frac{3SE}{4} - \frac{SE}{4} + \frac{2EI}{4} \\
&= \rho - \frac{4SE}{4} + \frac{2EI}{4} \\
&= \rho - SE + \frac{1}{2}EI \\
&= (2-E)\rho + \frac{E}{2}I \\
&= \underbrace{(2-E)\rho + \frac{E}{2}I}_{= \Sigma_2(\rho)}
\end{aligned}$$

$$\underline{\sum_{i=0}^3 K_i^\dagger \rho K_i = \Sigma_2(\rho)} \quad , \text{ there for } \underline{\Sigma_2(\rho) \text{ is CP}}$$

NB! Coefficient $2 - \frac{3E}{4}$ must be ≥ 0 for $0 \leq E \leq 1$

Then $\Sigma_2(\rho)$ is completely positive

$$\underline{\text{For } \Sigma_3(\rho) = \varepsilon \text{Tr}[\rho] |0\rangle\langle 0| + (2-E)\rho}$$

Choose operator:

$$K_0 = \sqrt{2-E} I$$

$$K_1 = \sqrt{E} |0\rangle\langle 0|$$

$$K_2 = \sqrt{E} |0\rangle\langle 1|$$

$$\text{Then, } \sum_{i=0}^2 K_i^\dagger \rho K_i = K_0^\dagger \rho K_0 + K_1^\dagger \rho K_1 + K_2^\dagger \rho K_2$$

$$\sum_{i=0}^2 K_i^* S K_i^+ = (I_d - E) S (I_d - E)^+ + E (107201) S (E (107201))^+ + E (107201) S (107201)^+ E$$

$$= (I_d - E) S + E \underbrace{(107201) S (107201)}_{= S_{00}} + E \underbrace{(107201) S (107201)}_{= S_{22}} E$$

$$= (I_d - E) S + E S_{00} (107201) + E S_{22} (107201)$$

$$= (I_d - E) S + E S_{00} (107201) + E S_{22} (107201)$$

$$= (I_d - E) S + E \underbrace{(S_{00} + S_{22})}_{= \text{Tr}[S]} (107201)$$

$$= \underbrace{(I_d - E) S + E \text{Tr}[S] (107201)}_{= \Sigma_3(S)}$$

$\sum_{i=0}^2 K_i^* S K_i^+ = \Sigma_3(S)$, Therefore, $\Sigma_3(S)$ is completely positive

For, C_P

All three $\Sigma_1(S)$, $\Sigma_2(S)$ and $\Sigma_3(S)$ are completely positive because they written in the form of

$$\boxed{\sum_{i=0}^2 K_i^* S K_i^+ = \Sigma_i(S)}$$

For linearity of quantum map

$$\text{for } \begin{cases} p_i \geq 0 \\ \sum_i p_i = 1 \end{cases}$$

$$\sum (\sum p_i \rho_i) = \sum p_i \sum \rho_i$$

$$\rho = \sum_i p_i \rho_i$$

But we have to verify all three
of $\sum \rho_i$, $E(\rho)$, $\sum \rho_i$
one by one

Then,

$$\sum \rho_i = (\lambda - E)\rho + E X \rho X$$

Take,

$$\sum \rho_i (\sum p_i \rho_i) = (\lambda - E) \sum p_i \rho_i + E X (\sum p_i \rho_i) X$$

$$= \sum p_i \left((\lambda - E) \rho_i + E X \rho_i X \right) = \sum \rho_i$$

~~Therefore,~~

$$= \sum p_i \sum \rho_i$$

linear

$$\text{For, } \sum \rho_i = (\lambda - E)\rho + \frac{E}{2} \text{Tr}[\rho] I$$

$$\sum \rho_i (\sum p_i \rho_i) = (\lambda - E) \sum p_i \rho_i + \frac{E}{2} \text{Tr} [\sum p_i \rho_i] I$$

$$= (\lambda - E) \sum p_i \rho_i + \frac{E}{2} \sum p_i \text{Tr} [\rho_i] I$$

$$= \sum_i p_i \left(\underbrace{(\lambda - \epsilon) \rho_i + \frac{\epsilon}{2} \text{Tr}[\rho_i] I}}_{\Sigma_2(\rho_i)} \right)$$

$$= \underline{\underline{\sum_i p_i \Sigma_2(\rho_i)}}, \text{ linear.}$$

For $\Sigma_3(\rho) = (\lambda - \epsilon) \rho + \epsilon \text{Tr}[\rho] |0\rangle\langle 0|$

Then $\Sigma_3(\sum_i p_i \rho_i) = (\lambda - \epsilon) \sum_i p_i \rho_i + \epsilon \text{Tr}[\sum_i p_i \rho_i] |0\rangle\langle 0|$

$$= (\lambda - \epsilon) \sum_i p_i \rho_i + \epsilon \sum_i p_i \text{Tr}[\rho_i] |0\rangle\langle 0|$$

$$= (\lambda - \epsilon) \sum_i p_i \rho_i + \epsilon \sum_i p_i \text{Tr}[\rho_i] |0\rangle\langle 0|$$

$$= \sum_i p_i \left(\underbrace{(\lambda - \epsilon) \rho_i + \epsilon \text{Tr}[\rho_i] |0\rangle\langle 0|}_{= \Sigma_3(\rho_i)} \right)$$

$$= \underline{\underline{\sum_i p_i \Sigma_3(\rho_i)}}, \text{ linear}$$

There for all three noise channel are linear,

Conclusion

All three are valid quantum operations (CPTP)
maps